

Did My Model Predict the World Cup, or Remember It?

An audit of a fine-tuned LLM for football prediction
Assignment 2 (Mini Project): Predicting a Sport Game

Zanwen Fu

September 2026

Relationship to prior work. The data pipeline, fine-tuned model, and prediction files used here were built for my IDS 598.1 final project in Spring 2026 ([spring report](#)). This assignment does not resubmit that project. It *audits* it: every test, baseline, figure, and conclusion in Sections 4–7 is new, and several of them overturn claims I made in the spring. Code: <https://github.com/zanwenfu/football-llm>. AI use is disclosed in Section 9.

Summary

In the spring I fine-tuned Llama 3.1 8B on the 2010–2018 World Cups and reported that, before kickoff, it called over/under 2.5 goals correctly for 76.6% of 2022 World Cup matches, more than 20 points ahead of a tabular XGBoost model, and that it turned \$1,000 into \$15,684 in a simulated backtest. For this assignment I asked a harder question: *how much of that was prediction?* Llama 3.1’s pretraining data runs to December 2023, so the 2022 results were already in the model’s weights before I fine-tuned it.

Three tests say the headline was mostly memory. Hiding the team names cuts exact-score accuracy from 43.8% to 10.9%. With names shown, 59% of predictions are the real scoreline or its home/away mirror, against about 10% by chance, and the tournament’s famous upsets come back with the right digits and the wrong winner. And the same backtest pays +1,125% to a rule with no model at all. Once the model cannot tell which match it is looking at, it is statistically indistinguishable from betting “under” every time. What survives is that the halftime score carries real information, but a one-line rule already captures it.

My conclusion is not that LLMs cannot predict football. It is that **accuracy on past matches is not evidence of skill for a model that may have read the results**, and that the test which could show skill is a forward test on matches the model has never seen, priced against real odds.

1 The question, and what would prove me wrong

Target question. Can a large language model predict football matches well enough to be worth pursuing as a betting strategy?

My starting point was simple: I wanted to see whether an LLM can predict matches correctly, and whether this is a promising track. I believed it was. An LLM has read a great deal about football, it can take player statistics as text, and it can be conditioned on the state of a game at halftime without retraining. I expected that with more data and a stronger model it could produce positive returns over the long run.

The spring project tested this on two decisions a bettor actually makes: who wins (1X2) and whether the match goes over or under 2.5 goals (O/U), each made either before kickoff or at halftime.

What would falsify it. If the model has real skill, that skill should not depend on it knowing *which* match it is predicting. If its advantage disappears once it cannot identify the match, then what I measured was recall of results it had read, not prediction. I built this control in the spring (a copy of every test prompt with the team names replaced by “Team A” and “Team B”) but, as Section 4 shows, I read it on the wrong metric.

2 Analytical framework

Following the research process from Lecture 5, I sampled the questions that different roles would ask of this project, and then checked which of their measures my data can support.

Role	Key question	Measure	Available to me?
Bettor / trader	Does the model beat the price after the bookmaker's cut?	Return against real odds; break-even hit rate (52.6% at odds 1.90)	No odds data. The spring backtest assumed 1.90 on every match.
Match analyst	Who is stronger, and what is the game state?	Starting XI's prior-season player stats; halftime score	Yes (API-Football)
Sports statistician	How many goals should each side score?	Dixon-Coles expected goals λ, μ ; O/U probability	Only World Cup results (256 matches)
ML evaluator	Is the measured skill real, or leaked?	Named vs. hidden gap; mirror test; trivial baselines; power	Yes (built for this audit)

Table 1. Roles, key questions, measures, and data availability.

Lecture 7 ranks the data types that decide soccer predictions. Setting my inputs against that ranking was sobering (Table 2): the model saw only the fifth-ranked type and none of the top four.

Rank	Data type (Lecture 7)	In my model's input?
1	Historical match outcomes	Only World Cup matches, 3–7 per team every four years
2	Betting-market probabilities	None
3	Team strength ratings (Elo, etc.)	None
4	Team performance statistics (xG, form)	None (StatsBomb xG exists for 2018/2022 only; unused)
5	Player statistics & availability	Yes: the only real input
8	Match context (venue, stage)	Yes, which turns out to matter for leakage (Section 4.3)

Table 2. The model's inputs against the lecture's ranking of predictive data types.

3 What I built, and what I claimed

Pipeline. From API-Football I scraped all 256 World Cup matches from 2010 to 2022 (results, half-time scores, lineups, events) and about 168,000 rows of player statistics (one per player, season and competition) from 2004 to 2025. For each match, the starting XI's statistics from the three seasons *before* the tournament are aggregated into a short text profile per team. Llama 3.1 8B Instruct was fine-tuned with QLoRA on the 192 matches of 2010, 2014 and 2018 to answer Prediction / Score / Reasoning, and tested on the 64 matches of 2022, each shown twice: with team names, and with names hidden.

Models compared in this audit (all on the same 64 matches): the LLM with team names; the LLM with names hidden; XGBoost trained on the same features; a Dixon-Coles model fitted to 2010–2018 results (new, following Lecture 7); and rules that use no model (“always under”, “double the halftime goals”).

What the spring report claimed. LLM and XGBoost tied on 1X2, but the LLM won on goals: 39.1% vs. 6.2% exact scores and 76.6% vs. 54.7% on O/U before kickoff. I attributed this to “pretrained scoreline priors” and called it a magnitude/direction decomposition. I also reported that halftime conditioning lifted 1X2 accuracy with $p = 0.024$, that adding first-half events raised O/U accuracy to 79.7%, and a simulated return of +1,468%. Several of those numbers do not survive the audit, and one cannot be regenerated at all (Section 4.5).

4 The audit

Lecture 6 gives a seven-item validation checklist. I worked through it in the order that turned out to matter.

4.1 Out-of-sample test: the test set was not out of sample

Llama 3.1 was released in July 2024 with pretraining data through December 2023. The 2022 World Cup, including every score, was on the public web the model was trained on. My split was clean for the *fine-tuning* data (2010–2018 in, 2022 out) and for the features (only seasons before each tournament), but not for the *base model*. This is exactly the failure in Lecture 7's W-5 case: “training data holds post-event knowledge”, and “pretend not to know” does not work because the knowledge is in the parameters.

4.2 Cross-validation with outcomes: hide the names

Knowing the names does not help the model pick the winner at all (50.0% vs. 53.1%), but it quadruples exact scores. In the spring my notebook compared the two versions on 1X2 only, saw no gap, and concluded the model “learned from stats, not names”. **I had the right control and read it on the one metric where memory helps least.** A model that remembers “it finished 1–0” but not who won is exactly what this table looks like.

64 matches, same prompts except names	Pregame			Halftime		
	1X2	Exact	O/U 2.5	1X2	Exact	O/U 2.5
LLM, team names	50.0	43.8	78.1	64.1	51.6	81.2
LLM, names hidden	53.1	10.9	56.2	64.1	18.8	70.3
Paired McNemar p (names vs. hidden)	0.82	<0.001	0.007	1.00	<0.001	0.12

Table 3. Accuracy (%) with and without team names. The model is identical; only the names differ.

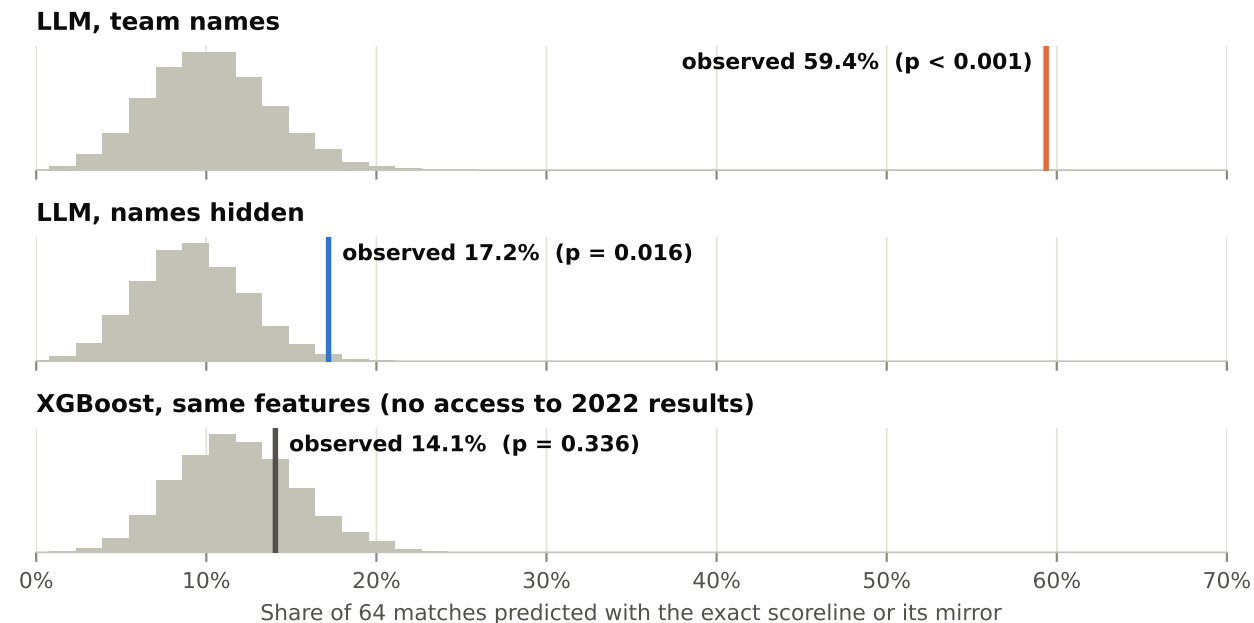


Figure 1. Exact-or-mirrored scorelines, pregame. Gray: 10,000 shuffles of each model’s own predictions. XGBoost, which has no route to the 2022 results, sits inside its null.

4.3 Model interpretation: the mirror test

If the model half-remembers a result, it should often reproduce the scoreline’s digits but not always their order. I counted predictions that match the real score exactly *or* with home and away swapped, and compared that rate with a null in which each model’s own predictions are matched to shuffled fixtures. The null keeps both the model’s score distribution and the tournament’s, so it already credits common scores like 1–0.

With names, 59.4% of predictions are the real score or its mirror ($p < 0.001$). The ten mirrored cases are telling: *Tunisia 1–0 France* was predicted 0–1, *Morocco 1–0 Portugal* 0–1, *Japan 2–1 Germany* as *Germany 2–1*, *Qatar 0–2 Ecuador* 2–0. The model recalls the scoreline and assigns the win to the stronger team on paper. It also got scores no statistics could produce, such as England 6–2 Iran, Portugal 6–1 Switzerland, and 3–3 for the final, which is the score *after extra time*. This single mechanism explains the “magnitude/direction decomposition” I reported in the spring: it is the signature of memory, not a scoreline prior.

The hidden-name model also beats its null ($p = 0.016$) while XGBoost on the same features does not ($p = 0.34$). The hidden prompts still contain the coach’s name, the stadium and the round

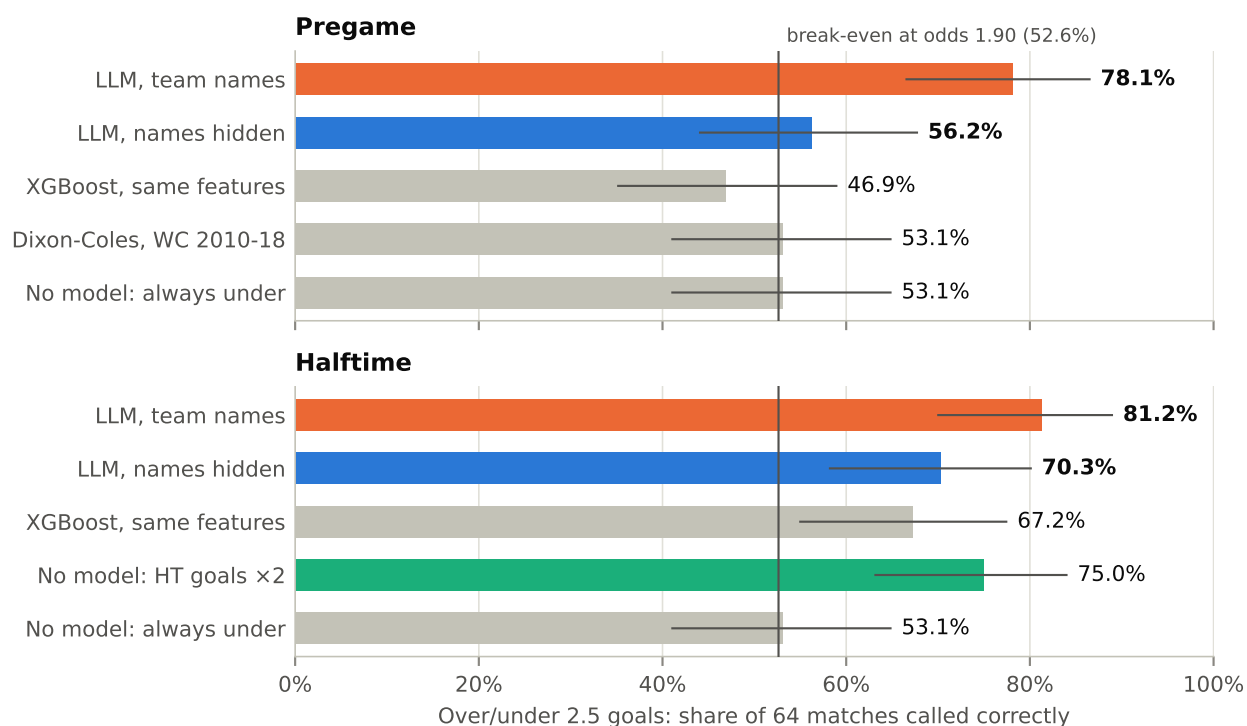


Figure 2. O/U 2.5 accuracy on the same 64 matches, with 95% Wilson intervals. The vertical line is the hit rate needed to break even at odds of 1.90.

(“Team A | Coach: G. Southgate”, “Group Stage – 1 | Al Bayt Stadium”), which together identify the match. The evidence is weak at $n = 64$, but it means even my “names hidden” numbers are an upper bound.

4.4 Cross-validation with benchmarks

Model (64 matches)	Pregame			Halftime	
	1X2	Exact	O/U	1X2	O/U
LLM, names hidden	53.1	10.9	56.2	64.1	70.3
XGBoost, same features	57.8	12.5	46.9	62.5	67.2
Dixon-Coles, fitted on WC 2010–2018	45.3	7.8	53.1	–	–
No model: always home win / always under	45.3	–	53.1	45.3	53.1
No model: halftime leader wins / halftime goals $\times 2$	–	–	–	56.2	75.0

Table 4. Every model on the same matches (%). With names hidden, nothing clearly beats a constant.

Pregame, with the names hidden, the LLM’s 56.2% on O/U is within noise of “always under” (53.1%); XGBoost is worse than that constant. At halftime, simply doubling the first-half goals gets 75.0%, ahead of the hidden-name LLM’s 70.3% (paired $p = 0.66$).

The Dixon-Coles model from Lecture 7 taught me something on its own. I chose its shrinkage by fitting on 2010–2014 and scoring 2018, never looking at 2022, and the best out-of-sample setting

was maximum shrinkage: every team average. Team strengths from one World Cup barely transfer to the next, because each team plays three to seven matches every four years. With this data the textbook benchmark collapses to a constant, which is Lecture 7's own warning ("Not yet, the current model is overfitting with the data set!") and says the real bottleneck is data type 1 in Table 2.

4.5 Data validation

- **A result I cannot reproduce.** The spring report's third regime (halftime plus first-half events: 79.7% O/U, calibration error 0.182, $p = 0.006$) came from a scratch run whose predictions were never saved. I drop those numbers.
- **Labels include extra time.** Bets settle at 90 minutes, but my labels use the final score (Argentina–France is labelled 3–3; it was 2–2 at 90 minutes). This affects 2 test and 9 training matches. The Dixon-Coles model trains on 90-minute scores.
- **The hidden prompts are only partly hidden** (coach, stadium, round; above).

4.6 Stability

- **Two runs, two answers.** Inference sampled at temperature 0.1. Two runs of the same weights on the same prompts gave 52.3% and 51.6% 1X2 accuracy, and my spring README mixed numbers from both.
- **Double counting.** The spring $p = 0.024$ for the halftime lift pooled the named and hidden copies ($n = 128$), counting every match twice. Tested within each copy ($n = 64$), the lift is $p = 0.049$ with names and $p = 0.21$ without.
- **Width of the evidence.** At $n = 64$ a 95% interval on an accuracy is about ± 12 points wide, so most model-to-model differences here are not resolvable.

4.7 Error analysis

The outputs have problems that are about decoding, not modelling. Before kickoff the model predicted 1 draw in 64 matches with names and 0 without, against 15 real draws. It also produced 9 (names) and 11 (hidden) scorelines with one side on six or more goals, such as 0–8, 2–8 and 6–8, against 3 real ones. And in 24 of 64 pregame outputs its written label contradicts its own score. Inference used a repetition penalty of 1.1, which discourages repeating a digit and may be why 0–0 and 1–1 come out as 0–8 and 1–6. I have not verified this, since it needs a GPU rerun.

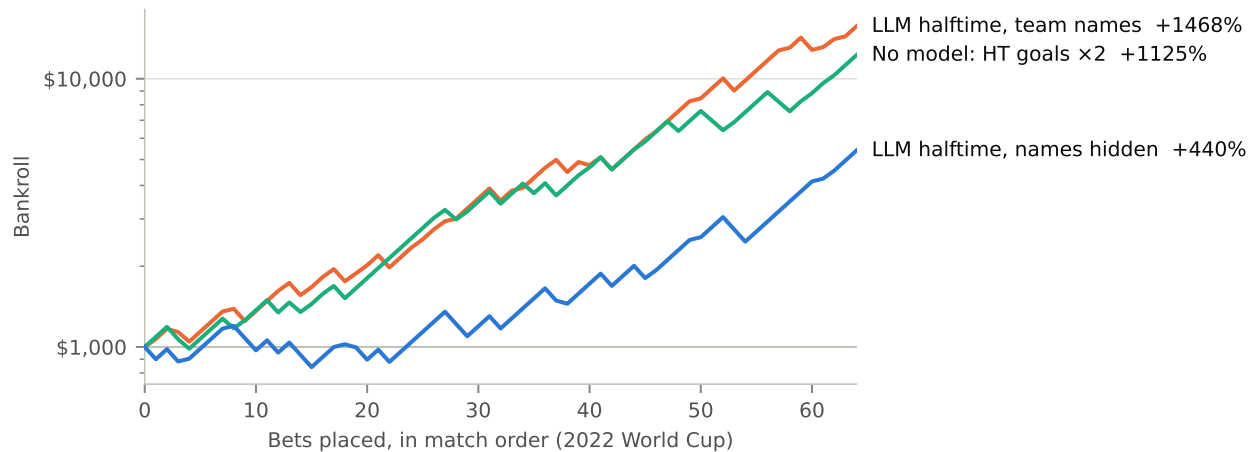


Figure 3. The spring backtest (O/U 2.5 at odds 1.90 on every match, quarter Kelly, 10% cap), run on three strategies. Log scale; \$1,000 start.

4.8 Backtest and forward test

The spring simulation assumed odds of 1.90 on both sides of every match. At halftime that assumption hands an edge to anyone who can see the score, because a real bookmaker reprices the line when it is 2–0 at the break. The proof is Figure 3: **a rule with no model returns +1,125%**. The hidden-name LLM returns +440% at halftime and -0.4% before kickoff. The simulation was measuring memory and its own odds assumption, not skill.

Even with real odds, one tournament cannot establish an edge. With flat stakes at odds 1.90, detecting a true edge of 5% per bet (one-sided $\alpha = 0.05$, power 0.8) takes about 2,200 bets; 2% takes about 13,900. A World Cup has 64 matches, 104 from 2026.

5 What survives

- **The halftime score is real information.** With names hidden, halftime conditioning moves O/U from 56.2% to 70.3% and 1X2 from 53.1% to 64.1%, though neither is significant at $n = 64$ ($p = 0.09$, $p = 0.21$). The model generalises to a prompt field it was never trained on. But a one-line rule captures the same information at least as well.
- **Before kickoff, I find no skill** once the model cannot identify the match.
- **The infrastructure is sound and reusable:** temporal feature windows, the named/hidden control, and now an audit that runs in one second and flags leakage automatically. That is what makes the forward test in Section 7 cheap.

6 What I believe now

Then: an LLM can predict matches; with more data and a stronger model it would produce positive long-run returns.

Now: the evaluation was the bottleneck, not the model. Four things changed my mind about how to get there:

1. **A stronger model would make this backtest look better for the wrong reason.** A newer, larger LLM has read more results. On any test set before its training cutoff it will score higher by remembering more. Scaling the model makes the measurement problem worse unless the test is on matches it cannot have seen.
2. **Capacity cannot create information that is not in the input.** My prompts held data type 5 and none of types 1–4. A bigger model reads the same player totals.
3. **Returns come from beating prices, not from accuracy.** A 56% hit rate is only valuable against a price that implies less, and I never compared to a real price.
4. **World Cups are too small to prove an edge.** The “long run” needs thousands of bets, which means club leagues, where markets are most efficient.

I still think the idea is worth testing, in narrower forms: an LLM as a reader of unstructured information the market prices slowly (team news, injuries, confirmed lineups), especially in thinner markets; or as a feature extractor feeding a calibrated model such as Dixon-Coles, rather than as the predictor itself. **What would change my mind** is a pre-registered forward test on matches after the model’s training cutoff, with identities properly hidden, that beats closing odds by more than the bookmaker’s margin over hundreds of bets.

7 Plan for the final project

Gap found in this audit	Fix
Test set inside the base model’s training data	Forward test on matches after December 2023: Euro 2024 and Copa América 2024 (about 83 matches) and the 2026 World Cup (104 matches)
Hidden prompts leak via coach, stadium, round	Strip them; confirm with the mirror test
No market benchmark	Collect closing odds; report return and closing-line value
Sparse history	All international results for Dixon-Coles and Elo benchmarks
Decoding problems	No repetition penalty, deterministic decoding, several samples per match for probabilities
Extra-time labels	Train and score on 90-minute results
Halftime prompts never trained on	Fine-tune on halftime prompts; compare against the halftime rule

Table 5. From Assignment 2 to the final project.

8 Reproducing every number

No GPU is needed; everything runs in seconds from the committed prediction files.

```
pip install -e ".[dev]"
python scripts/leakage_audit.py --figures-dir report/figures # Sec. 4-5
python -m football_llm.baselines.dixon-coles               # benchmark
python scripts/reproduce_paper.py --skip-figures             # spring tables
```

9 AI use disclosure

I used AI assistants throughout, and I am disclosing all of it as the assignment requires.

- **Spring project.** Code for the data pipeline, training, evaluation and serving, and drafting of the spring report and README, were done with heavy AI assistance.
- **This assignment.** Claude (Anthropic) reviewed the repository and flagged the contamination risk (the first dialogue in AI_USAGE.md). In a Claude Code session it then wrote the audit and Dixon-Coles code and their tests, ran the analyses, and drafted this report and the updated README from my direction. I chose the project and the question, and the conclusions in Section 6 are mine. Every number in this report is printed by the scripts in Section 8.

Dialogues. Both are attached in full in AI_USAGE.md at the root of the repository: Claude's replies from the conversation that reviewed the project, and the Claude Code session that built the audit and drafted this report (my prompts, Claude's replies, and a one-line summary of every action it took).